

CausalCache: Fixed-Budget Visual Memory Reallocation for Frozen GUI Agents

Anonymous submission

Abstract

Long-horizon GUI agents retain their full interaction trace cheaply as text, but can afford to show the policy only a few past events in pixels. We formulate this as *fixed-budget visual memory reallocation*: every event survives in summary form, an active budget B caps how many regain their archived screenshot, and the only decision is which B events those are. The ubiquitous Recent- B heuristic spends every slot on the latest events; CAUSALCACHE instead deploys a budget-aware set selector—trained offline from measured set utilities and attached to a *completely frozen* policy—that recomposes the whole window when distant evidence outranks a recent frame. Zero-shot on a 117-task mobile benchmark whose memory-critical split is partitioned mechanically from the benchmark’s own metadata, reallocation lifts overall success from 30.2% to 36.8% ($p=0.035$) and memory-critical success from 19.4% to 30.6% ($p=0.006$), with no detectable movement on single-app controls. Three content- and composition-matched controls locate the mechanism: substituting each restored screenshot with the turn’s own verbatim text erases the entire gain, OCR text recovers only part of it, and capping the selector to single-slot replacement collapses success exactly to the Recent- B level—the gain is carried by *pixels*, restored in *multi-slot compositions*, placed where distant visual dependency exists. On desktop, restoring history to high fidelity is worth about 13 success points over summary-only memory. The token-gated adapter used to measure set utilities offline is behaviorally neutral at deployment, so the entire gain is delivered without touching a single policy weight.

Introduction

Autonomous agents that follow natural-language instructions and operate graphical user interfaces promise broad practical value: they can automate repetitive digital chores, carry out multi-application workflows on behalf of their users, and extend what a single person can accomplish on desktop and mobile devices (Xie et al. 2024; Rawles et al. 2025; Xu et al. 2026). Delivering that value on long tasks turns memory into a first-order design problem: GUI agents accumulate screenshots, actions, text arguments, coordinates, and interface changes over many decisions. A complete action trace persists cheaply as compressed text, but keeping

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

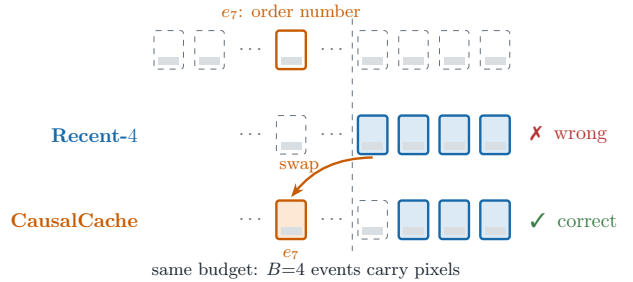


Figure 1: Fixed-budget reallocation in a cross-app episode ($B=4$). All events remain as text. Recent- B assigns pixels to the latest four events and misses distant evidence e_7 ; CAUSALCACHE swaps e_7 into the same four-slot visual budget and produces the correct order number.

every screenshot active is costly and distracting. Each past event therefore has two policy-visible fidelities: a summary-only record and, when needed, that same record with its archived screenshot reattached. The memory-control question is not only which events remain recorded, but which recorded events should be exposed again in high-fidelity pixels (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026).

This paper treats the visual window as an allocation problem. Under a fixed budget B , every past event persists as text and exactly B events regain their archived screenshot. Recent- B —the default in existing controllers, which retain recent events, retrieve visually similar screenshots, or learn salience from task supervision (Zeng et al. 2026; Shi et al. 2026; Liu et al. 2026)—is one point in this allocation space; CAUSALCACHE searches the rest of it. The system is deliberately minimal at deployment: the policy stays *completely frozen*—no fine-tuning, no adapters active, no extra weights—and a lightweight budget-aware selector (two-tower marginal scorer, beam composition, exact- B with recent fallback) recomposes the visual window at each step, with a conditional second decoding pass that lets the policy re-decide under the recomposed window. Figure 1 illustrates the reallocation on a real cross-application episode.

Our contributions are three. (1) **A deployment-free re-allocation system.** Attached to a frozen GUI-Owl-1.5-8B policy, the selector transfers zero-shot to a 117-task mo-

mobile benchmark it never trained on, lifting overall success by +6.6pp ($p=0.035$) and memory-critical success by +11.3pp ($p=0.006$) over Recent- B at matched budget, with single-app controls unchanged. **(2) A mechanism, isolated by construction.** Content-matched controls show the carrier must be pixels (verbatim-text restoration collapses to Recent- B ; OCR text recovers only part), composition-capped controls show multi-slot recomposition is necessary (a $k \leq 1$ cap collapses exactly to Recent- B), and a metadata-partitioned split plus a budget sweep show the gain sits precisely where evidence lies beyond the recent window. **(3) An offline utility-measurement recipe.** Set utilities are measured on desktop traces with a token-gated K/V probe (HGKV) trained under difference-in-differences supervision inside a pre-specified drift envelope; the selector distills these measurements. The probe’s role is *teaching*, never shipping: at deployment it is behaviorally neutral and the distilled selector carries the entire gain; a selector distilled from frozen-policy-scored labels recovers most of the closed-loop margin but trails the HGKV-taught one on the memory-critical split.

Related Work

Visual GUI agents and benchmarks. WebShop, WebArena, Mind2Web, WorkArena, WorkArena++, VisualWebArena, and REAL span grounded shopping, real websites, enterprise workflows, and visual web interaction (Yao et al. 2022; Zhou et al. 2024; Deng et al. 2023; Drouin et al. 2024; Boisvert et al. 2024; Koh et al. 2024; Garg et al. 2025). AITW, AndroidControl, AndroidWorld, OSWorld, and GUIOdyssey extend executable evaluation to mobile and desktop interfaces (Rawles et al. 2023; Li et al. 2024a; Rawles et al. 2025; Xie et al. 2024; Lu et al. 2025). Screenshot-native agents learn grounding and actions directly from pixels (Hong et al. 2024; Cheng et al. 2024; Lin et al. 2025; Xu et al. 2026). These works measure long-horizon interaction; our question is which completed event should re-enter a bounded prompt in pixels. MobileWorld further stresses long-horizon, cross-application mobile workflows under reproducible functional evaluation (Kong et al. 2026).

Agent memory and reusable experience. RAG retrieves external memory, ReAct interleaves reasoning with action, and later agents preserve reflections, experience, or evolving long-term stores (Lewis et al. 2020; Yao et al. 2023; Shinn et al. 2023; Zhao et al. 2024; Zhong et al. 2024; Gutiérrez et al. 2024). ICAL retrieves multimodal programs of thought (Sarch et al. 2024). These systems write or retrieve artifacts across interactions or tasks; CAUSALCACHE instead holds a within-episode summary trace fixed and controls whether an existing event is exposed as summary only or summary plus image.

Long-horizon GUI memory. GUIOdyssey resamples prior screenshots, MementoGUI learns memory operators, and AndroTMem retrieves causally linked anchor states (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026). Their primary variable is which item is retained or retrieved. CAUSALCACHE keeps the compressed trace fixed, controls event fidelity conditioned on the already-active visual context, and trains the interface that consumes restored pixels.

Context and visual-token compression. Prompt compressors remove textual redundancy (Jiang et al. 2023, 2024; Pan et al. 2024); KV methods retain, evict, select, or quantize cache states (Xiao et al. 2024; Zhang et al. 2023; Liu et al. 2023; Li et al. 2024b; Liu et al. 2024; Corallo and Papotti 2024); and visual methods merge or prune tokens within an image (Bolya et al. 2023; Chen et al. 2024; Yang et al. 2025). Their unit is a token or KV position. CAUSALCACHE instead preserves the summary trace and allocates whole archived screenshots across events; HGKV controls only how promoted history-image tokens are consumed.

Interventional memory value. Recent work studies memory through controlled interventions or active state influence (Srivastava 2026; Liu et al. 2026). Our intervention is narrower and executable: a selected event regains one archived post-action image while the remaining prompt stays fixed, yielding a policy-specific behavioral surrogate rather than an environment-level causal effect. Set models parameterize coalition-conditioned scores (Zaheer et al. 2017; Lee et al. 2019), whose marginals connect to cooperative-game attribution (Shapley 1953); we use controlled restorations to train a deployable selector for one fixed GUI policy.

Method

Problem: Fixed-Budget Reallocation

At step t the agent holds a complete textual trace of events $1, \dots, t-1$ (instruction, per-step summaries, verbatim responses) and an archive of their screenshots. The active context C_t renders every event as text and promotes a subset S_t of past events to summary-plus-screenshot form, subject to the budget

$$|S_t| = \min(B, t-1), \quad (1)$$

so every allocation renders exactly the same number of images and any promotion of a distant event *displaces* a recent one. Restoration moves an official retained turn—verbatim response plus screenshot—as a unit; no pixels are ever reconstructed from text. Let $Q(S)$ be the policy’s mean target-token log-likelihood of the successful next action when events in S are exposed in summary-plus-image form. Recent- B fixes $S_t = S_{\text{recent}}$, the B most recent distinct events; CAUSALCACHE targets

$$S^* = \arg \max_{\substack{S \subseteq \{1, \dots, t-1\} \\ |S| = \min(B, t-1)}} Q(S), \quad Q(S^*) > Q(S_{\text{recent}}), \quad (2)$$

so recency is one candidate allocation, not a separate modality. Given a partial allocation S , conditional marginals $\Delta_t(j | S) = Q(S \cup \{j\}) - Q(S)$ drive selection: a distant event enters only by beating the marginal of the recent realization it evicts (“causal” denotes policy-specific effects of matched prompt interventions, not structural causal discovery). Every reported set utility re-scores the complete restored set; the realized replacement intensity $k = |S \setminus \text{Recent-}B|$ is a measured statistic, never preset.

Method

CAUSALCACHE has two parts: an offline utility-measurement stage that produces training labels, and a deployed selector

that recomposes windows for a frozen policy.

Measuring set utility offline. On desktop traces we build matched-budget replacement groups: at a decision point with target action a_t^* , every prompt carries exactly B history images and differs only in which event holds pixels:

- **Recent:** Recent- B ;
- **Relevant restore:** the oldest recent slot replaced by the target-specific archived screenshot (age $\geq B+2$, strictly older than the window);
- **Wrong restore:** the same slot replaced by an age-matched screenshot whose following action is not equivalent to a_t^* .

Utility is the teacher-forced log-likelihood margin of a_t^* between the recomposed and recent windows. To make these margins sensitive, we instrument the policy with HGKV, a token-gated K/V adapter on the last eight layers that touches only restored history-image tokens and is bitwise bypassed when no history image is present. Let ℓ denote mean target log-likelihood, superscripted by adapter state (active or frozen bypass); each arm is anchored to *its own* frozen score:

$$\begin{aligned} A_s &= \ell_{\text{relevant}}^{\text{on}} - \ell_{\text{relevant}}^{\text{off}}, & A_r &= \ell_{\text{recent}}^{\text{on}} - \ell_{\text{recent}}^{\text{off}}, \\ A_n &= \ell_{\text{wrong}}^{\text{on}} - \ell_{\text{wrong}}^{\text{off}}, \end{aligned} \quad (3)$$

and the DiD objective is

$$\begin{aligned} \mathcal{L} &= [m - (A_s - A_r)]_+ + [m - A_s]_+ \\ &\quad + [m - (A_s - A_n)]_+ \\ &\quad + \lambda_{\text{cap}}([|A_r| - \epsilon]_+ + [|A_n| - \epsilon]_+) + \lambda \|\Delta W\|_2^2, \end{aligned} \quad (4)$$

with $m=0.01$, $\epsilon=0.02$, $\lambda_{\text{cap}}=2$, $\lambda=10^{-4}$, and no action cross-entropy. On the identity adapter every increment is zero, so uniform amplification of any history cancels exactly; only *selective* use of target-relevant content reduces the loss. HGKV passes pre-specified selectivity gates that the frozen policy and an ungated control do not match (Table 3). Crucially, HGKV is an offline *instrument*: deployment does not require it (Table 2).

Why single-slot replacement ($k=1$). Training teaches only the $k=1$ replacement interface: offline mining finds simultaneous multi-frame demand rare, and a matched-budget probe shows the first replacement is worth its slot while the paired increments of a second and third are indistinguishable from zero or negative. Larger- k allocations remain expressible at deployment because the selector composes exact- B sets with recent fallback, so k stays a result statistic; mining details and the full probe are in the supplement.

Set-level labels, not singletons. Because deployment composes whole windows, labels must cover multi-slot compositions. A beam teacher expands scored sets from singleton seeds (shortlist 6, beam 3) up to the full budget, scoring every edge $U(S \cup \{j\}) - U(S)$ by re-scoring the entire composed window; the resulting 147k set edges supervise marginals conditioned on non-empty partial sets $|S| \in \{1, 2, 3\}$ alongside 58k singleton edges. This is what licenses the deployed selector’s multi-slot recompositions—at $B=4$ it replaces the entire recent window on 26% of states—and the $k \leq 1$ -capped

control shows exactly these compositions carry the closed-loop gain.

Deployed selector. A two-tower marginal scorer $\hat{\Delta}(j | S)$ (cheap candidate features plus set-context features; recency-membership features carry substantial weight) drives a beam-3 composition to an exact- B set with recent fallback. A conditional second pass re-decodes the action under the recomposed window and always replaces the first proposal—decoding is greedy everywhere, so the second pass cannot act as best-of- n . The selector adds under 40 ms per step; the second pass costs $\sim 10\%$ of per-step wall time at $B=4$ and less at larger budgets.

Reference actions. Part of the selector’s input asks whether the action taken right after an archived event matches a reference action. Training uses the gold teacher-forced target, an offline oracle unavailable at deployment; CAUSALCACHE therefore re-grounds the reference in the policy’s own tentative action: the policy first acts on the Recent- B prompt, and re-decides on the re-allocated prompt only when a recent visual realization is displaced. This proposal reference is statistically indistinguishable from the gold oracle on held-out desktop groups, whereas weaker references—no action signal, or instruction-only similarity—degrade selection and over-replace; a single-pass variant reusing the previously executed action removes the second call but transfers worse across platforms (reference ladder and efficiency analysis in the supplement).

Experiments

Experimental Setup

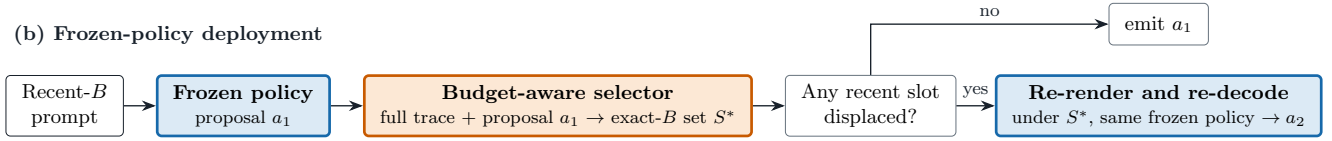
Training Data and Model Selection. Policy adaptation and selector training use only desktop data: successful AgentNet/OpenCUA Ubuntu trajectories (5,000 screened, 2,293 successful, 6,003 decision points) yield fixed-budget replacement groups at $B=1/2/4$ of 969/764/470 (2,203 in total; 1,774 training units, 211 development groups, and 218 held-out test groups, trajectory-disjoint), complemented by a high-precision OSWorld witness seed mined from official successful trajectories. A further 160 groups at $B=8$ are constructed but deliberately excluded from training, so $B=8$ evaluates out-of-training budget extrapolation. Checkpoints and thresholds are selected only on the desktop development split under pre-specified gates: positive DiD selection with a positive bootstrap lower bound, and drift caps $|A_r|, |A_n| < \epsilon$. No mobile benchmark data touches training or selection. All adapter training, offline scoring, and closed-loop serving run on NVIDIA H200 Tensor Core GPUs.

The Fixed-Budget Fidelity-Restoration Design. Figure 2 shows the two learned components and the same-budget re-decision that defines the core comparison. Every compared allocation carries exactly B history images with identical resolutions, summaries, and prompt structure, so the contrast isolates *which summarized events are promoted to high fidelity*. $B=4$ is the primary deployment-like setting; the realized number of non-recent promotions k is measured rather than preset.

(a) Offline utility distillation



(b) Frozen-policy deployment



Only the selector is deployed; all policy weights remain frozen.

Figure 2: Training-to-deployment pipeline. Offline, complete matched-budget windows are scored to derive set-conditioned marginal labels for the budget-aware selector. At deployment, the frozen policy proposes a_1 under Recent- B ; the prompt is re-rendered and re-decoded only if the selector returns an allocation different from Recent- B .

Policies and Fidelity-Allocation Baselines. The frozen policy is GUI-Owl-1.5-8B-Instruct (Xu et al. 2026). Under identical data, steps, and cadence we compare the frozen policy, an ungated KV control (the same last-eight layers, K/V targets, rank, and alpha as HGKV but active on all tokens), and HGKV itself (token-gated, structurally bypassed at $B=0$). Fidelity-allocation baselines are Recent- B , Random, OCR/RGB similarity, and the learned budget-aware restoration selector (exact- B , recent fallback); a beam oracle over the candidate pool bounds headroom. Every method receives the same summaries and the same active image budget.

Zero-Shot Benchmarks and Metrics. The policy interface and selector, trained on desktop data only, are evaluated (i) on held-out desktop decision groups (offline, per- B), (ii) zero-shot on an uncontaminated OSWorld task roster (Xie et al. 2024) with official evaluator scores (tasks whose trajectories seeded the witness corpus are excluded from zero-shot claims and reported separately as in-domain diagnostics), and (iii) zero-shot on a cross-app mobile memory benchmark of 117 runnable tasks (62 cross-app memory candidates vs. 55 single-app controls, split mechanically by the benchmark’s task metadata), which contributes no data to any training or selection decision (Kong et al. 2026). Offline metrics are teacher-forced log-likelihood margins and full-action equivalence under an argument and coordinate tolerance (sensitivity settings in the supplement); closed-loop metrics are paired task success with cluster bootstrap, step counts, and wall time. All arms decode greedily, so every action is a deterministic function of its rendered prompt, and the conditional second pass always replaces the proposal rather than selecting between samples—it cannot act as best-of- n .

Results

Main result. Table 1 gives the headline: at matched budget, learned reallocation on the frozen policy lifts the full mobile roster by +6.6pp and the memory-critical stratum by +11.3pp, with controls unmoved. Table 2 decomposes the system: summary-only, Recent- B , and adapter-on-Recent- B

Benchmark / stratum	Recent- B	CAUSALCACHE	Δ
MobileWorld full (117)	30.2	36.8	+6.6*
memory-critical (62)	19.4	30.6	+11.3*
single-app control (55)	42.4	43.6	+1.2
split $\times$ method interact.	—	—	+10.4*
OSWorld-Verified (361)	32.7	34.1	+1.1

Table 1: Primary zero-shot contrasts at deployment budget $B=4$: frozen policy + selector vs. frozen policy + Recent- B (task-paired bootstrap; *the 95% CI excludes zero, $p=0.035$ full / $p=0.006$ memory-critical; interaction two-sided $p=0.022$). Recent- B : three-run means; selector arm: single round. The OSWorld frozen+selector contrast is +1.1pp (paired CI $[-2.5, +4.7]$, n.s.); the adapter-attached variant scores 32.7 there, likewise indistinguishable from Recent- B .

are pairwise indistinguishable (the behavioral face of the drift cap), while both selector arms separate from all of them and not from each other. The gain is a property of *which events hold pixels*, not of any weight update. An outcome-defined check agrees: on the 95 tasks the summary-only baseline cannot solve, reallocation gains 7.0pp, and it adds nothing on the 22 it already solves. A frozen-policy recent-dose sweep makes the recency alternative explicit: across $B=0-4$, success is non-monotone with every paired contrast against $B=0$ crossing zero—promoting additional recent events alone buys nothing.

Mechanism I: the carrier must be pixels. Two controls keep selection, budget, and the two-pass interface intact but replace each restored frame’s pixels with text: the turn’s verbatim recorded response, or an OCR transcription of the same screenshot. On the memory-critical split verbatim text collapses to the Recent- B level (18.0% vs. 19.4%; -10.4 pp below pixels, task-paired $p=0.006$), ruling out a generic “reminder” effect; OCR recovers part of the level (24.2%) but its +4.8pp margin over Recent- B does not separate from

Method	OSWorld-Verified (361)	MobileWorld (117)	MW memory-critical (62)	MW control (55)
Frozen, summary-only ($B=0$)	19.9	28.2	21.0	36.4
Frozen + Recent- B	32.7	29.9	22.6	38.2
HGKV + Recent- B	33.0	30.2	19.4	42.4
HGKV + selector	32.7	33.9	28.0	40.6
Frozen + selector (CAUSALCACHÉ)	34.1	36.8	30.6	43.6
frozen-taught selector variant	—	33.3	25.8	41.8

Table 2: Closed-loop deployment decomposition at $B=4$. OSWorld-Verified: official no-GDrive roster, official 15-step budget. MobileWorld: zero-shot; three-run means for multi-run arms, single rounds otherwise. The three Recent/summary arms are pairwise indistinguishable, and the two selector arms are pairwise indistinguishable (+2.9pp full, CI $[-2.9, +8.6]$): allocation carries the gain, the measurement adapter is deployment-neutral, and the best deployment ships zero adapted weights. Swapping the label teacher from HGKV to the frozen policy keeps a significant margin over Recent- B (+6.5pp memory-critical) but cedes part of it—the instrument earns its keep at training time. On desktop, high-fidelity memory is worth +12.8pp over summary-only ($19.9 \rightarrow 32.7$) while allocation variants are indistinguishable there—consistent with placement: evidence rarely falls outside the recent window when state is re-observable.

zero, while pixel restoration’s +11.3pp does. Full tables in the supplement.

Mechanism II: multi-slot recomposition is necessary. Capping the deployed selector at one replacement per step ($k \leq 1$; dedup-aliased recent tails count as kept) collapses memory-critical success to exactly the Recent- B level (19.4% vs. 19.4%; -8.6 pp below the uncapped adapter-attached system, $p=0.02$). At $B=4$ the uncapped selector keeps the full recent window on 19% of states and replaces the whole window on 26%: those multi-slot compositions are not system completeness—they carry the gain. Although the HGKV teacher’s policy corpus supervises only the $k=1$ replacement interface, the selector’s set-level labels cover multi-slot compositions by construction, and the closed loop confirms the composed allocations transfer.

Mechanism III: the gain sits where the evidence sits. The memory-critical split is defined by the benchmark’s own metadata, before any measurement: cross-app tasks force evidence beyond the recent window. The split $\times$ method interaction is +10.4pp (two-sided $p=0.022$). Figure 3 visualizes the contrast; Figure 4 extends the closed loop along the budget axis ($B \in \{1, 2, 4, 8\}$, same frozen weights): the allocation gain is positive at every budget, pooled to +4.1pp ($p=0.004$; task-paired bootstrap), and on the memory-critical subset the margin is largest at the trained budgets and narrows at the untrained $B=8$, where the eight-frame recent window already reaches back far enough to cover part of the split’s evidence. (Budget-sweep arms carry the deployment-neutral measurement adapter; per-budget tables in the supplement.)

The offline instrument. Table 3 reports the pre-specified desktop DiD gate at $B=1$ on 94 development groups. On this split the frozen policy shows no reliable preference for a task-relevant archived screenshot over the informative recent frame it would displace: to the frozen policy the two are worth about the same, so profitable fidelity reallocation must be learned—and measured with an instrument that can see it. HGKV passes every gate with a selectivity margin the layer- and rank-matched ungated control does not reach, while holding recent-window and wrong-history

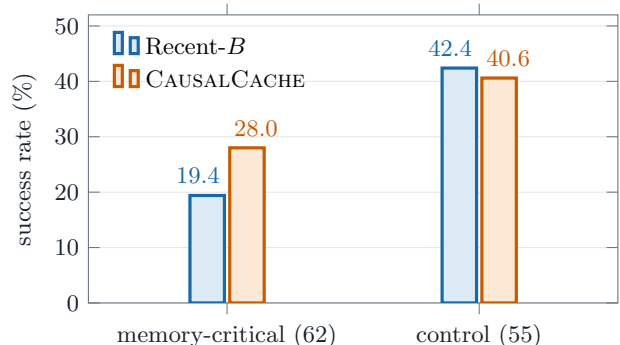


Figure 3: Memory-critical vs. single-app control success at $B=4$: +8.6pp on the memory-critical stratum, no detectable movement on controls, interaction $p=0.022$.

drift well inside the cap. The same recipe transfers across backbone scale untouched: on GUI-Owl-1.5-32B it passes every pre-specified gate with DiD select +0.0142 (95% CI $[+0.011, +0.018]$) and recent-window drift within a third of the cap. The closed-loop transfer does not follow: mounting the 8B-taught selector on a frozen 32B policy scores 39.2 against 49.0 for Recent-4 (-9.8 pp, paired $n=51$, CI $[-21.6, +2.0]$; coverage caveat in the supplement). The 32B Recent-4 baseline is itself 19pp above 8B’s, so a larger backbone already resolves from the recent window much of what 8B needs archived pixels for; an 8B probe labels 8B’s deficits, not 32B’s. Selector transfer across backbone scale therefore requires relabeling with the deployed policy, even though the probe recipe itself transfers untouched.

Selectivity Is Not Inherited from the Frozen Policy. On the desktop development split, the frozen policy shows no reliable preference for a task-relevant archived screenshot over the informative recent frame it would displace (recent-baseline construction details in the supplement): to the frozen policy the two are worth about the same, so profitable fidelity reallocation must be learned. Under the official multi-turn protocol the frozen replacement effect decays with the win-

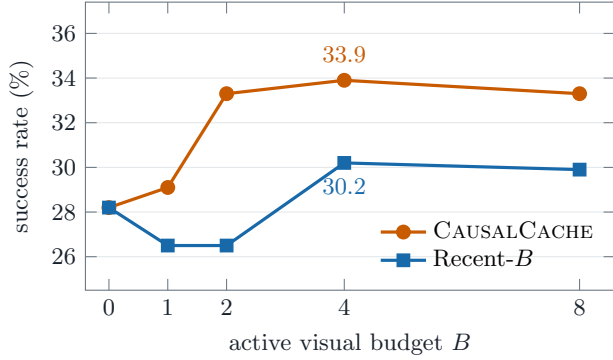


Figure 4: Closed-loop MobileWorld success along the budget axis (full roster). At $B=0$ the two arms are identical; $B=0/4$ are three-run means, other budgets single rounds. Pooled over B : $+4.1\text{pp}$, $p=0.004$; per-budget table in the supplement.

dow and reverses at large B (supplement), while the learned interface keeps a positive selection effect across the budget axis.

Selective Use Within a Drift Envelope. Table 3 reports the pre-specified desktop DiD gate at $B=1$. Adapters are trained for matched steps; checkpoints follow the frozen selection rule. HGKV passes every gate: its selected checkpoint learns a positive DiD selection effect against the redundant-duplicate recent control, a re-scoring probe confirms the effect against the informative true-previous-frame control, and recent-only and wrong-history drift stay well inside the cap $\epsilon=0.02$; its no-history path is bitwise identical to the frozen policy under randomized nonzero adapter weights. The layer- and rank-matched ungated control also learns selectivity, confirming that the last-eight K/V position carries signal, but token gating adds a significant paired per-group advantage while holding drift further from the caps. Gating is what buys selectivity *inside* the envelope.

Fidelity Reallocation and Zero-Shot Transfer. The offline budget axis shows the same picture. Across the trained budgets $B \in \{1, 2, 4\}$, HGKV maintains positive selectivity—per- B DiD gates all pass with drift within the pre-specified caps—and the learned allocation improves over Recent- B at matched image counts, while keeping the full recent window in 19% of $B=4$ states. The ungated control again passes but with uniformly smaller selection effects, drifting toward the cap at the largest budget. Untrained-budget extrapolation ($B=8$) and the full per-budget tables are in the supplement.

Policy-side per- B gates and the frozen recent-dose sweep appear in Results. Table 4 brackets the learned selector between allocation rules that share its budget and rendering exactly. Random exact- B and RGB-similarity restoration are indistinguishable from Recent- B : sparse exposure of archived pixels, or visual similarity to the current screen, is not utility. Inverting the learned score is significantly harmful at both budgets—a specificity control showing the scorer ranks real signal symmetrically rather than exploiting a ren-

Adapter (desktop DiD, $B=1$)	DiD select	Wrong drift
HGKV (gated)	+0.0224	0.0085
Ungated KV control	+0.0177	0.0071

Table 3: Token-gated K/V probe on 94 development groups. Both adapters hold drift inside the pre-specified cap (0.02); gating buys a significantly larger selectivity margin per group. The probe generates training labels and never ships. On GUI-Owl-1.5-32B the same recipe passes all gates (DiD select +0.0142, CI strictly positive).

Allocation rule	$B=2$	$B=4$
Learned scorer	+0.030	+0.017
Random exact- B	+0.006	+0.008
RGB similarity	+0.006	+0.007
Inverted score	−0.016	−0.009
No recency feats	+0.012	+0.008

Table 4: Chooser ladder: log-likelihood margin over Recent- B at matched budgets with fresh full-set policy scoring. Only the learned ranking and its inversion separate from zero (episode-clustered intervals in the supplement).

dering artifact. Retraining without the recency-membership features returns the selector to the noise band, so budget-context features carry substantial weight.

The budget is spent adaptively: at $B=4$ the learned scorer keeps the entire recent window on 19% of states and replaces all of it on 26%—a state-dependent k profile that neither random allocation nor RGB similarity reproduces. Although training supervises only the $k=1$ interface, these multi-slot compositions sustain the closed-loop gains of Table 2: the token-gated adaptation applies per restored frame and generalizes across replacement multiplicity.

Deployment cost of the second pass. The conditional second pass fires on 81–87% of audited mobile steps; the selector costs under 40 ms. Instrumented across budgets, the added model time per step is 2.0/2.3/2.9 s at $B=1/2/4$ —8.1/8.7/10.5% of the measured per-step wall time—and falls to roughly 5% at $B=8$, where the second pass fires on only 39% of steps. On desktop the median end-to-end task overhead is 7%. The re-decision’s real-time cost thus peaks near one-tenth of wall time at the primary budget; complete intervals are in the supplement.

Desktop closed loop. On OSWorld-Verified (361 tasks, official no-GDrive roster and 15-step budget), all high-fidelity arms reach 32.7–33.0%, each about +13pp over summary-only memory (19.9%, paired 95% CIs excluding zero), while same-budget allocations remain indistinguishable. OSWorld therefore supports high-fidelity *availability*; conditional *allocation* is tested by the mobile memory-critical split. HGKV again shows no detectable behavioral difference on matched Recent- B allocations, mirroring the mobile finding. CAUSAL-CACHE adds a median of eight policy calls per task, about 7% of median end-to-end task time.

Extended-horizon diagnostic (OSWorld). Under the standard 15-step setting, OSWorld provides limited exposure to fidelity reallocation: 352/361 tasks run to the step cap, and within 15 steps the evidence rarely leaves the recent window. We therefore include a matched 30-step horizon analysis (Recent- B vs. frozen+selector, both full-roster rounds) to test whether allocation gains emerge as relevant evidence moves farther beyond the recent window. They do not, but establishing that required first removing a hardware confound that our initial analysis missed and then rerunning the affected episodes.

Episodes were sharded across whichever machines were free, and the machines are not interchangeable: three were H200 hosts, one was an older A6000 host. On the 30-step round the A6000 host returned 26.4% where the H200 hosts returned 33.9–36.9%. The deficit is not a task-mix artifact—it survives direct standardization to a common domain mix (27.9%)—and it is not merely slower execution: on the *same* episodes the A6000 host and an H200 rerun both take a median of 30 steps and both exhaust the cap on 50% of episodes, yet differ by 7.3pp in success. It concentrates in graphics-heavy applications (GIMP 10.0 vs. 41.2, Impress 6.7 vs. 34.4) while text and navigation domains are unaffected, the signature of a renderer slow enough that screenshots are captured before the interface settles.

That host contributed 110 of the selector arm’s 361 episodes and none of the baseline’s. We reran exactly those 110 on H200 hosts and substituted them, leaving every other episode untouched. The selector arm rises 2.2pp and the 30-step contrast moves from -2.8pp to -0.5pp (CI $[-4.7, +3.6]$): on matched hardware the two allocations are indistinguishable at 30 steps, and the earlier reading that the selector’s 15-step edge collapses under horizon extension was an artifact of one machine.

Across the full roster the selector arm reads 34.1/35.5/34.1/35.2% at 15/30/50/100 steps—a $6.7\times$ range of step budget spanned by 1.4 percentage points, with every horizon contrast crossing zero. Cap exhaustion falls only from 42.7% to 36.6% over the same range, so the residual failures are not step-starved. The official 15-step cap therefore does not suppress the baseline in our favor, and the selector’s small 15-step edge is neither amplified nor destroyed by horizon extension—consistent with OSWorld evidence rarely leaving the recent window even at 100 steps.

Sharding also lets us separate run-to-run noise from the hardware effect, because some episodes were executed twice. Between two H200 hosts, 106 duplicated episodes agreed 98 times with balanced disagreements (5 versus 3), and a second, independent set of 22 duplicates from the rerun flipped twice: single-round noise on identical hardware is 7–9% of episodes. Between mismatched hardware the same measurement gives 7 flips out of 10. That order-of-magnitude gap is the clearest single indication that what we removed was a hardware artifact rather than ordinary variance. The main 15-step comparison is unaffected: neither of its arms touched the A6000 host, and the two H200 hosts it does use are the matched pair characterized above.

Limitations and Conclusion

The evidence assembles into three steps. First, the problem is fidelity allocation under a fixed visual budget: every event survives as text and only B regain pixels. Second, the frozen policy cannot measure which distant frame deserves a slot; the HGKV probe learns exactly that selectivity inside a drift envelope that leaves matched-recent behavior untouched, and its measurements teach the selector. Third, the learned selector reallocates profitably exactly where distant visual dependency exists: memory-critical tasks improve, controls show no detectable change. The headline selector arm is a single round against three-run baselines. The frozen-teacher variant carries 60% of the HGKV-taught selector’s set labels (88.8K against 147.3K scored set edges), so its mobile gap confounds teacher identity with label volume. Volume does bind at the low end: at 9% of the states every selector we trained—either teacher, paired with either policy—falls below Recent-4 (supplement), so the factorial isolating teacher identity is only interpretable in the volume regime where selectors work at all. We ran the same comparison on OSWorld but do not report a verdict from it: the two arms were executed on disjoint sets of machines, so no episode is paired across the comparison, and the reading flips sign ($+0.6$ to -1.7pp) depending on which version of the selector arm it is taken against. Since we have paired evidence that two hosts can differ and no paired evidence about this particular pair, the desktop teacher contrast is uninterpretable as run. Whether the mobile teaching advantage generalizes to desktop is therefore open, and settling it requires rerunning one arm on the other’s machines. OSWorld shows fidelity value but no allocation headroom at any horizon we tested (15/30/50 steps), so the reallocation claim is mobile-only. We control an active context budget, not archive capacity, and CAUSALCACHE reattaches rather than generates pixels; the content-matched controls attribute the gain to the pixels themselves. We claim only that some summarized events have greater conditional marginal utility than the recent realization they displace; pre-training contamination remains possible. In one sentence: under a fixed visual budget, a frozen GUI agent’s memory is best spent by *recomposing* the window—several slots at a time, in pixels—exactly where distant evidence outranks recency.

References

- Boisvert, L.; Thakkar, M.; Gasse, M.; Caccia, M.; Le Selliier de Chezelles, T.; Cappart, Q.; Chapados, N.; Lacoste, A.; and Drouin, A. 2024. WorkArena++: Towards Compositional Planning and Reasoning-Based Common Knowledge Work Tasks. In *Advances in Neural Information Processing Systems*, volume 37.
- Bolya, D.; Fu, C.-Y.; Dai, X.; Zhang, P.; Feichtenhofer, C.; and Hoffman, J. 2023. Token Merging: Your ViT but Faster. In *The Eleventh International Conference on Learning Representations*.
- Chen, L.; Zhao, H.; Liu, T.; Bai, S.; Lin, J.; Zhou, C.; and Chang, B. 2024. An Image Is Worth 1/2 Tokens After Layer 2: Plug-and-Play Inference Acceleration for Large Vision-Language Models. In *Computer Vision – ECCV 2024*.

- Cheng, K.; Sun, Q.; Chu, Y.; Xu, F.; YanTao, L.; Zhang, J.; and Wu, Z. 2024. SeeClick: Harnessing GUI Grounding for Advanced Visual GUI Agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 9313–9332. Bangkok, Thailand: Association for Computational Linguistics.
- Corallo, G.; and Papotti, P. 2024. FINCH: Prompt-Guided Key-Value Cache Compression for Large Language Models. *Transactions of the Association for Computational Linguistics*, 12: 1517–1532.
- Deng, X.; Gu, Y.; Zheng, B.; Chen, S.; Stevens, S.; Wang, B.; Sun, H.; and Su, Y. 2023. Mind2Web: Towards a Generalist Agent for the Web. In *Advances in Neural Information Processing Systems*, volume 36. Datasets and Benchmarks Track.
- Drouin, A.; Gasse, M.; Caccia, M.; Laradji, I. H.; Del Verme, M.; Marty, T.; Vazquez, D.; Chapados, N.; and Lacoste, A. 2024. WorkArena: How Capable Are Web Agents at Solving Common Knowledge Work Tasks? In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 11642–11662. PMLR.
- Garg, D.; Caples, D.; Draguns, A.; Ravi, N.; Putta, P.; Garg, N.; Hebbar, P.; Joo, Y.; Gu, J.; London, C.; Schroeder de Witt, C.; and Motwani, S. 2025. REAL: Benchmarking Autonomous Agents on Deterministic Simulations of Real Websites. In *Advances in Neural Information Processing Systems*, volume 38.
- Gutiérrez, B. J.; Shu, Y.; Gu, Y.; Yasunaga, M.; and Su, Y. 2024. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. In *Advances in Neural Information Processing Systems*, volume 37.
- Hong, W.; Wang, W.; Lv, Q.; Xu, J.; Yu, W.; Ji, J.; Wang, Y.; Wang, Z.; Dong, Y.; Ding, M.; and Tang, J. 2024. CogAgent: A Visual Language Model for GUI Agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 14281–14290.
- Jiang, H.; Wu, Q.; Lin, C.-Y.; Yang, Y.; and Qiu, L. 2023. LLMingua: Compressing Prompts for Accelerated Inference of Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 13358–13376.
- Jiang, H.; Wu, Q.; Luo, X.; Li, D.; Lin, C.-Y.; Yang, Y.; and Qiu, L. 2024. LongLLMingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 1658–1677.
- Koh, J. Y.; Lo, R.; Jang, L.; Duvvur, V.; Lim, M.; Huang, P.-Y.; Neubig, G.; Zhou, S.; Salakhutdinov, R.; and Fried, D. 2024. VisualWebArena: Evaluating Multimodal Agents on Realistic Visual Web Tasks. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 881–905. Bangkok, Thailand: Association for Computational Linguistics.
- Kong, Q.; Zhang, X.; Yang, Z.; Gao, N.; Liu, C.; Tong, P.; Cai, C.; Zhou, H.; Zhang, J.; Chen, L.; Liu, Z.; Hoi, S.; and Wang, Y. 2026. MobileWorld: Benchmarking Autonomous Mobile Agents in Agent-User Interactive and MCP-Augmented Environments. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 6142–6167. San Diego, California, United States: Association for Computational Linguistics.
- Lee, J.; Lee, Y.; Kim, J.; Kosiorek, A. R.; Choi, S.; and Teh, Y. W. 2019. Set Transformer: A Framework for Attention-Based Permutation-Invariant Neural Networks. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 3744–3753. PMLR.
- Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.-t.; Rocktäschel, T.; Riedel, S.; and Kiela, D. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, volume 33, 9459–9474.
- Li, W.; Bishop, W.; Li, A.; Rawles, C.; Campbell-Ajala, F.; Tyamagundlu, D.; and Riva, O. 2024a. On the Effects of Data Scale on UI Control Agents. In *Advances in Neural Information Processing Systems*, volume 37.
- Li, Y.; Huang, Y.; Yang, B.; Venkitesh, B.; Locatelli, A.; Ye, H.; Cai, T.; Lewis, P.; and Chen, D. 2024b. SnapKV: LLM Knows What You Are Looking for Before Generation. In *Advances in Neural Information Processing Systems*, volume 37.
- Lin, K. Q.; Li, L.; Gao, D.; Yang, Z.; Wu, S.; Bai, Z.; Lei, S. W.; Wang, L.; and Shou, M. Z. 2025. ShowUI: One Vision-Language-Action Model for GUI Visual Agent. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 19498–19508.
- Liu, C.; Chen, L.; Zhou, H.; Zhang, X.; Kong, Q.; Tong, P.; Wang, W.; Yu, X.; Hoi, S.; and Wang, Y. 2026. What Memory Do GUI Agents Really Need? From Passive Records to Active Task-Driving States. arXiv:2606.31612.
- Liu, Z.; Desai, A.; Liao, F.; Wang, W.; Xie, V.; Xu, Z.; Kyrillidis, A.; and Shrivastava, A. 2023. Scissorhands: Exploiting the Persistence of Importance Hypothesis for LLM KV Cache Compression at Test Time. In *Advances in Neural Information Processing Systems*, volume 36.
- Liu, Z.; Yuan, J.; Jin, H.; Zhong, S.; Xu, Z.; Braverman, V.; Chen, B.; and Hu, X. 2024. KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 32332–32344. PMLR.
- Lu, Q.; Shao, W.; Liu, Z.; Du, L.; Meng, F.; Li, B.; Chen, B.; Huang, S.; Zhang, K.; and Luo, P. 2025. GUIOdyssey: A Comprehensive Dataset for Cross-App GUI Navigation on Mobile Devices. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 22404–22414.
- Pan, Z.; Wu, Q.; Jiang, H.; Xia, M.; Luo, X.; Zhang, J.; Lin, Q.; Rühle, V.; Yang, Y.; Lin, C.-Y.; Zhao, H. V.; Qiu, L.; and Zhang, D. 2024. LLMingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression. In *Findings of the Association for Computational Linguistics: ACL 2024*, 963–981.

- Rawles, C.; Clinckemaiellie, S.; Chang, Y.; Waltz, J.; Lau, G.; Fair, M.; Li, A.; Bishop, W. E.; Li, W.; Campbell-Ajala, F.; Toyama, D. K.; Berry, R. J.; Tyamagundlu, D.; Lillicrap, T. P.; and Riva, O. 2025. AndroidWorld: A Dynamic Benchmarking Environment for Autonomous Agents. In *The Thirteenth International Conference on Learning Representations*.
- Rawles, C.; Li, A.; Rodriguez, D.; Riva, O.; and Lillicrap, T. 2023. AndroidInTheWild: A Large-Scale Dataset for Android Device Control. In *Advances in Neural Information Processing Systems*, volume 36.
- Sarch, G. H.; Jang, L.; Tarr, M. J.; Cohen, W. W.; Marino, K.; and Fragkiadaki, K. 2024. VLM Agents Generate Their Own Memories: Distilling Experience into Embodied Programs of Thought. In *Advances in Neural Information Processing Systems*, volume 37, 75942–75985.
- Shapley, L. S. 1953. A Value for n -Person Games. In Kuhn, H. W.; and Tucker, A. W., eds., *Contributions to the Theory of Games II*, 307–317. Princeton, New Jersey: Princeton University Press.
- Shi, Y.; Li, J.; Zhang, L.; Dongfang, Z.; Wu, B.; Tao, S.; Yan, Y.; Qin, C.; Liu, W.; Lin, Z.; Li, H.; Huang, Y.; Dai, S.; Hei, Y.; Ding, Y.; Li, X.; Wang, S.; Xu, C.; Liu, J.; Ma, X.; Zheng, Z.; Zhang, X.; Wang, B.; Yang, N.; Wu, J.; Tian, L.; Li, C.; and Hu, X. 2026. AndroTMem: From Interaction Trajectories to Anchored Memory in Long-Horizon GUI Agents. arXiv:2603.18429.
- Shinn, N.; Cassano, F.; Gopinath, A.; Narasimhan, K.; and Yao, S. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems*, volume 36.
- Srivastava, S. S. 2026. Causal Intervention-Based Memory Selection for Long-Horizon LLM Agents. arXiv:2605.17641.
- Xiao, G.; Tian, Y.; Chen, B.; Han, S.; and Lewis, M. 2024. Efficient Streaming Language Models with Attention Sinks. In *The Twelfth International Conference on Learning Representations*.
- Xie, T.; Zhang, D.; Chen, J.; Li, X.; Zhao, S.; Cao, R.; Hua, T. J.; Cheng, Z.; Shin, D.; Lei, F.; Liu, Y.; Xu, Y.; Zhou, S.; Savarese, S.; Xiong, C.; Zhong, V.; and Yu, T. 2024. OS-World: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. In *Advances in Neural Information Processing Systems*, volume 37. Datasets and Benchmarks Track.
- Xu, H.; Zhang, X.; Liu, H.; Wang, J.; Zhu, Z.; Zhou, S.; Hu, X.; Gao, F.; Cao, J.; Wang, Z.; Chen, Z.; Liao, J.; Zheng, Q.; Zeng, J.; Xu, Z.; Bai, S.; Lin, J.; Zhou, J.; and Yan, M. 2026. Mobile-Agent-v3.5: Multi-Platform Fundamental GUI Agents. arXiv:2602.16855.
- Yang, S.; Chen, Y.; Tian, Z.; Wang, C.; Li, J.; Yu, B.; and Jia, J. 2025. VisionZip: Longer Is Better but Not Necessary in Vision Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 19792–19802.
- Yao, S.; Chen, H.; Yang, J.; and Narasimhan, K. 2022. WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents. In *Advances in Neural Information Processing Systems*, volume 35.
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *The Eleventh International Conference on Learning Representations*.
- Zaheer, M.; Kottur, S.; Ravanbakhsh, S.; Póczos, B.; Salakhutdinov, R.; and Smola, A. J. 2017. Deep Sets. In *Advances in Neural Information Processing Systems*, volume 30, 3391–3401.
- Zeng, Z.; Hua, H.; Zou, B.; Cai, M.; Feris, R.; and Luo, J. 2026. MementoGUI: Learning Agentic Multimodal Memory Control for Long-Horizon GUI Agents. arXiv:2605.18652.
- Zhang, Z.; Sheng, Y.; Zhou, T.; Chen, T.; Zheng, L.; Cai, R.; Song, Z.; Tian, Y.; Ré, C.; Barrett, C.; Wang, Z.; and Chen, B. 2023. H2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. In *Advances in Neural Information Processing Systems*, volume 36.
- Zhao, A.; Huang, D.; Xu, Q.; Lin, M.; Liu, Y.-J.; and Huang, G. 2024. ExpeL: LLM Agents Are Experiential Learners. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 19632–19642.
- Zhong, W.; Guo, L.; Gao, Q.; Ye, H.; and Wang, Y. 2024. MemoryBank: Enhancing Large Language Models with Long-Term Memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 19724–19731.
- Zhou, S.; Xu, F. F.; Zhu, H.; Zhou, X.; Lo, R.; Sridhar, A.; Cheng, X.; Ou, T.; Bisk, Y.; Fried, D.; Alon, U.; and Neubig, G. 2024. WebArena: A Realistic Web Environment for Building Autonomous Agents. In *The Twelfth International Conference on Learning Representations*.